Christian Döring

Graduate Student

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Education

2023 – 2026 (April – March)
M.Sc. Electrical and Computer Engineering,
Technical University of Munich
Thesis Title: Representing Mesoscale Detail in Neural Materials

2019 – 2023 (October – March)
B.Sc. Electrical and Computer Engineering,
Technical University of Munich
Thesis Title: Evaluation of Differentiable Inverse Rendering using Multi-View RGB Data

2011 – 2019 (September – June)
Abitur (A-Levels), *Gymnasium Bruckmühl*

Publications

2026
Real-time Rendering with a Neural Irradiance Volume
Arno Coomans, Giacomo Nazzaro, Edoardo A. Dominici, Christian Döring, Floor Verhoeven, Konstantinos Vardis, Markus Steinberger
arXiv

2024
Real-time Neural Rendering of Dynamic Light Fields
Arno Coomans, Edoardo A. Dominici, Christian Döring, Joerg H. Mueller, Jozef Hladky, Markus Steinberger
Project   In Computer Graphics Formum (EG), 2024

Experience

2026 – 2026 (June – Aug)
Research Intern, *Disney Research* Zurich,
- Post render edits

2026 – 2026 (May – May)
Research Intern, *Realistic Graphics Laboratory*, *EPFL*
- Neural Materials

2025 – 2026 (September – March)
Masters Thesis, *Realistic Graphics Laboratory*, *EPFL*
- Representing Mesoscale Detail in Neural Materials

2025 – 2025 (April – August)
Research Intern, *NVIDIA* Zurich
- Function Level Caching for Dr.Jit and Mitsuba 3
- Hash Grid for Dr.Jit
- Differentiable Radio Frequency Modeling with Sionna RT

2024 – 2025 April April	**Research Working Student**, *Huawei Technologies* Munich ○ Development on Dr.Jit/Mitsuba 3 ○ Real-time Neural Rendering Research
2023 – 2024 August February	**Research Intern**, *Huawei Technologies* Munich ○ Real-time Neural Rendering Research
2021 – 2021 July August	**Embeded Systems Intern**, *Aurum GmbH* Munich ○ Developed NFC library for STM32 in C
2017 July	**Intern**, *Lauterbach GmbH*
2017 July	**Intern**, *Electronic Theater Controls (ETC)* Holzkirchen

Side Projects

Hephaestus, Just In Time Compiler (JIT) for Vulkan, inspired by Dr.Jit. Implemented with own render graph solution. Includes cooperative matrix multiplication (KHR) and a port of tiny-cuda-nn in GLSL.
Source

Vulkan Path Tracer, Path tracer written in Rust using the screen-13 library. It supports the Disney BSDF with Next Event Estimation.
Source

Mitsuba 3 Experiments, Implementation of forward and differentiable path tracing algorithms in Mitsuba 3, such as ReSTIR GI and Large Steps in Inverse Rendering

Skills

Programming:
- **C/C++**, Rust
- **CUDA**, Vulkan
- **Python**, Lua
- **LaTeX**, Typst

Languages:
- **German** (native)
- English (B2+/C1)